

Technical Analysis: TCP Header Architecture and Wrap Around Dynamics

Summary

TCP is a **Transport Layer, connection-oriented, byte-stream protocol**. Its header and sequence management are fundamental to reliable delivery, flow control, and high-speed network integrity.

Key insights:

- TCP headers are **20–60 bytes**, controlled by a **4-bit Header Length (HLEN)** field with a $\times 4$ scaling factor.
- Sequence numbers track **every byte**, not just segments.
- **Wrap Around Time (WAT)**: 32-bit sequence numbers can repeat quickly at high bandwidths, potentially corrupting delayed segments.
- **Timestamps** in the Options field mitigate wrap-around conflicts.

1. TCP Header Structure

1.1 Dimensions

Component	Size	Notes
Fixed Header	20 bytes	Mandatory 5 rows $\times$ 32 bits
Options (Variable)	0–40 bytes	e.g., Timestamps, NOP padding
Header Length (HLEN)	4 bits	Actual header size = $HLEN \times 4$

HLEN Range: 5 $\rightarrow$ 20 bytes, 15 $\rightarrow$ 60 bytes. Non-multiples of 4 use padding.

1.2 Core Fields

Field	Size	Purpose
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Source Port	16 bits	Identifies sender application
Destination Port	16 bits	Identifies receiver application
Sequence Number	32 bits	First byte in segment
Acknowledgment Number	32 bits	Next expected byte from sender
Window Size	16 bits	Receiver buffer space
Checksum	16 bits	Header + data error detection
Urgent Pointer	16 bits	Points to end of urgent data

Optional fields (like Time Stamp) extend functionality without modifying core fields.

2. Port Addressing

TCP uses **16-bit port numbers** (0–65535) for unique process identification:

1. **Well-Known Ports (0–1023)**: IANA-assigned, e.g., HTTP (80), FTP (20/21), SMTP (25)
2. **Registered Ports (1024–49151)**: Reserved for company applications
3. **Dynamic/Private Ports (49152–65535)**: OS-assigned ephemeral ports

3. Sequence & Acknowledgment Management

- **Byte Stream Protocol**: Each byte has a unique sequence number.
- **Initial Sequence Number (ISN)**: Randomized for security.
- **Cumulative Acknowledgment (ACK)**: ACK = last received byte + 1
- **Data Size Calculation**: Since TCP lacks a "data length" field, derive from IP layer:

[
 $\text{Data Size} = \text{IP Total Length} - \text{IP Header} - \text{TCP Header}$
]

4. Wrap Around Time (WAT)

4.1 Sequence Space

- 32-bit sequence field $\rightarrow 2^{32} \approx 4.29 \times 10^9$ bytes.
- Maximum unique sequence numbers before wrapping: 4 GB of data.

4.2 Bandwidth Impact

$$\text{WAT} = \frac{2^{32} \text{ bytes}}{\text{Bandwidth (bytes/sec)}}$$

Bandwidth	WAT
1 Mbps	~4,294 s
1 Gbps	~4.29 s

- **Risk:** $\text{WAT} < \text{Packet Life Time}$ $\rightarrow$ delayed segments may collide with new sequence numbers $\rightarrow$ corruption or duplicate rejection.

4.3 Wrap Around Prevention

- Minimum sequence numbers required:

$$\text{Min Seq Numbers} = \text{Life Time} \times \text{Bandwidth}$$

- Bits required:

$$\text{Bits} = \lceil \log_2(\text{Life Time} \times \text{Bandwidth}) \rceil$$

- Bandwidth must be converted to **bytes/sec**, Life Time in seconds (typically 180 s).

